

Abiodun Victor Alagbada

PhD Candidate in Applied Mathematics

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RESEARCH INTERESTS

Koopman operator theory and data-driven dynamical systems; structural health monitoring (SHM); system identification and time-series learning; physics-informed machine learning and hybrid physics–data frameworks; numerical methods for stiff ordinary differential equations; optimization; reduced-order modelling.

EDUCATION

PhD in Applied Mathematics (in progress)

Bauhaus-Universität Weimar, Germany — Institute of Structural Mechanics

Supervisor: Prof. Dr. Tom Lahmer

Topic: Koopman-operator learning and hybrid physics–data frameworks for structural health monitoring

Funding: DAAD Scholarship; PTDF Overseas Scholarship 2024 – present

Postgraduate Diploma in Education (PGDE)

University of Ilorin, Nigeria 2022 – 2023

MSc in Mathematics

University of Jos, Nigeria

Thesis: *A Class of Block Second Derivative Method for Solving Stiff Ordinary Differential Equations* (Supervisor: Prof. Kumleng Geoffrey) 2017 – 2018

BSc in Mathematics, Second Class Upper

University of Jos, Nigeria

Thesis: *The Convergence of Some Complex Power Series* (Supervisor: Dr. Silas Longwap) 2010 – 2014

PUBLICATIONS

- [1] **Alagbada, A. V.**, and Lahmer, T. (2026). Koopman Operator Methods in Structural Health Monitoring: A Systematic Review Towards Hybrid Physics–Data Frameworks. *Applied Sciences*, 16(7), 3392. [doi:10.3390/app16073392](https://doi.org/10.3390/app16073392)
- [2] **Alagbada, A.**, Iroko, T., and Tchonetec, S. (2026). Machine Learning: Theoretical Foundations, Practical Applications, Ethical Implications, and Future Prospects — Exploring Challenges and Applications in Emerging Technologies. In *Encyclopedia of Modern Artificial Intelligence* (10 Volumes). IGI Global. [Chapter 407565](#)
- [3] Iroko, T., **Alagbada, A.**, and Tchonetec, S. (2026). Neural Regime-Switching Model

with Markov-Informed Attention for Short-Horizon Equity Return Forecasting. *Preprints.org* (preprint).

RESEARCH EXPERIENCE

Doctoral Researcher, Institute of Structural Mechanics

Bauhaus-Universität Weimar, Germany

2024 – present

- Develop interpretable, data-driven dynamical models for vibration-based condition assessment and damage diagnosis, grounded in Koopman operator theory.
- Build system-identification and time-series learning pipelines in Python with emphasis on reproducibility, benchmarking, and research-to-deployment translation.
- Conducted a PRISMA-guided systematic review of 67 peer-reviewed studies (2010–2025) on Koopman-based methods in SHM, published in *Applied Sciences*.
- Serve as SCRUM Master for a web-based pipeline-orchestration project integrating domain logic for engineering workflows.

TEACHING EXPERIENCE

Online Mathematics Tutor, Preply and Upwork

2020 – present

- Personalized instruction and academic coaching for international learners; digital teaching workflows with Jupyter, $\LaTeX$, and computational visualization.

Mathematics and Computer Science Teacher

Omega High School, Jos, Nigeria

2018 – 2023

- Taught Mathematics, Advanced Mathematics, and Computer Science; led curriculum planning, assessment moderation, and student mentoring.

Mathematics and Physics Teacher

Corpus Christi Comprehensive College, Kwall, Jos, Nigeria

2014 – 2017

PROFESSIONAL EXPERIENCE

Full-Stack Developer / Backend Engineer (project-based)

2022 – present

- Designed and deployed web and AI-based academic tools with Python (Django, Flask) and JavaScript (React, Node.js); RESTful APIs and containerized pipelines with Docker and GitHub Actions.

HONORS AND AWARDS

DAAD Scholarship, German Academic Exchange Service

2025

PTDF Overseas Scholarship, Petroleum Technology Development Fund,
Nigeria

2023/2024

Bronze Medal, National Mathematics Competition for University Students

2013

CERTIFICATIONS

- Qualified Teacher Status (QTS), UK Department for Education (TRN: 2361519)
- Registered Teacher, Teachers' Registration Council of Nigeria (2020)
- Software Engineering, Backend Specialization, ALX Africa (2023–2025)
- Web Design Consultancy, Nova University (2023)
- Backend Web Development, Side Hustle (2022); Full-Stack Development, Opecstat Academy (2022)

TECHNICAL SKILLS

Programming	Python (advanced), MATLAB, C/C++, JavaScript, SQL
Scientific computing	NumPy, Pandas, scikit-learn, TensorFlow, Jupyter/Marimo
Software engineering	Django, Flask, React, Node.js, REST APIs
Tools	Git/GitHub (advanced), Docker, GitHub Actions, $\text{\LaTeX}$ (advanced)

PROFESSIONAL MEMBERSHIPS

American Mathematical Society (AMS) · Mathematics Association of Nigeria · Black in Mathematics, South Africa

LANGUAGES

English (native) · Yoruba (native) · German (A2–B1)

REFERENCES

Available upon request. Doctoral supervisor: Prof. Dr. Tom Lahmer, Institute of Structural Mechanics, Bauhaus-Universität Weimar.

Last updated: July 2026